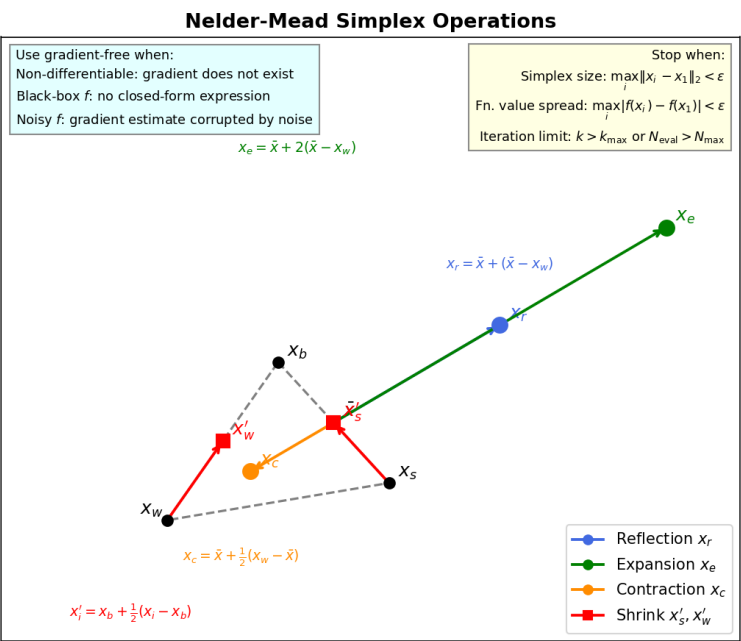




(h) [3 marks] The particle swarm update is:

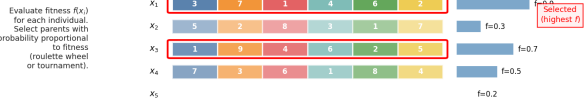
$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t, \quad v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}.$$

Define all symbols.

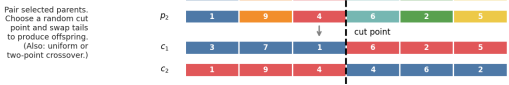
Solution:

- $x_k^{(i)}$: position of particle i at iteration k
- $v_k^{(i)}$: velocity of particle i at iteration k
- $x_{\text{best}}^{(i)}$: best position ever visited by particle i (personal best)
- x_{best} : best position ever found by *any* particle in the swarm (global best)
- α : inertia weight (momentum of previous velocity)
- β : cognitive coefficient (attraction to personal best)
- γ : social coefficient (attraction to global best)
- Δt : time step; k : iteration index; i : particle index

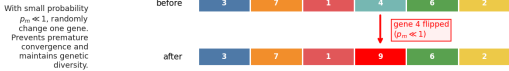
1. Selection



2. Crossover

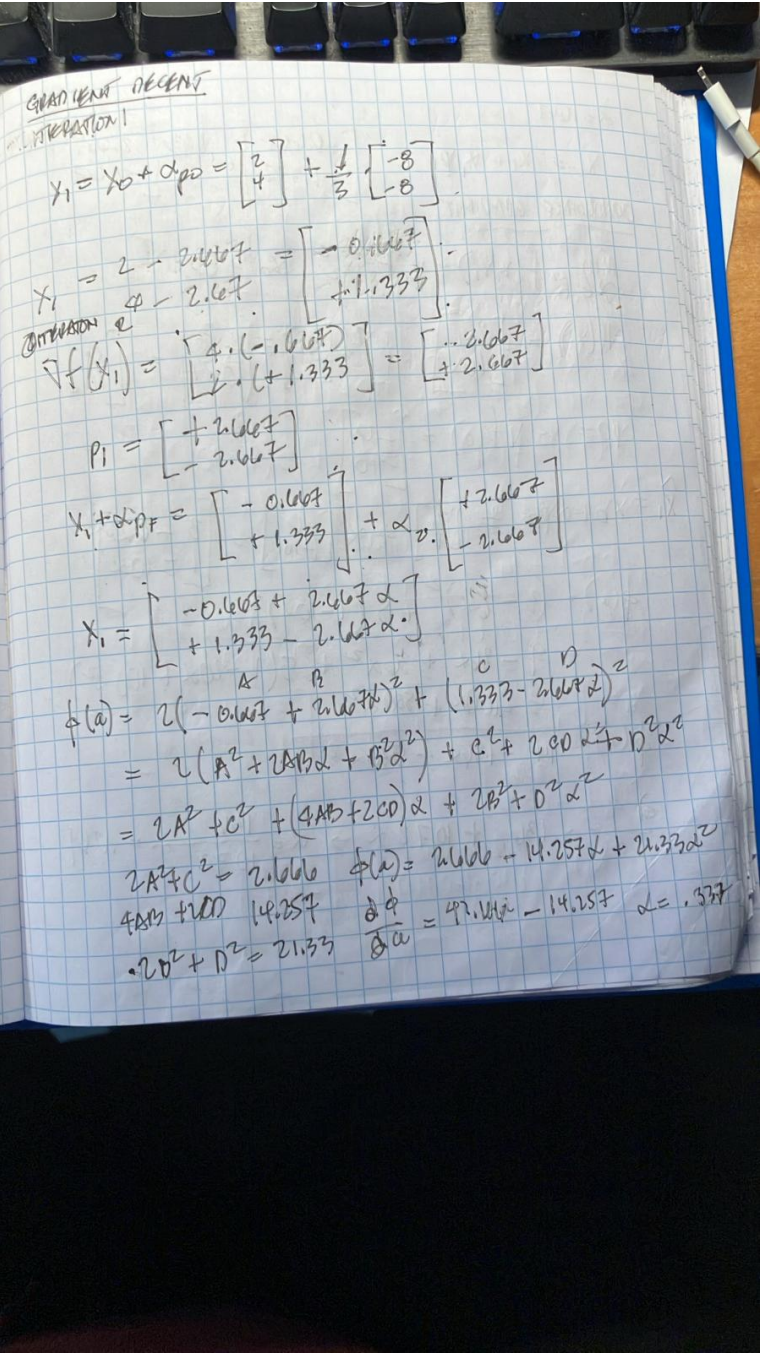
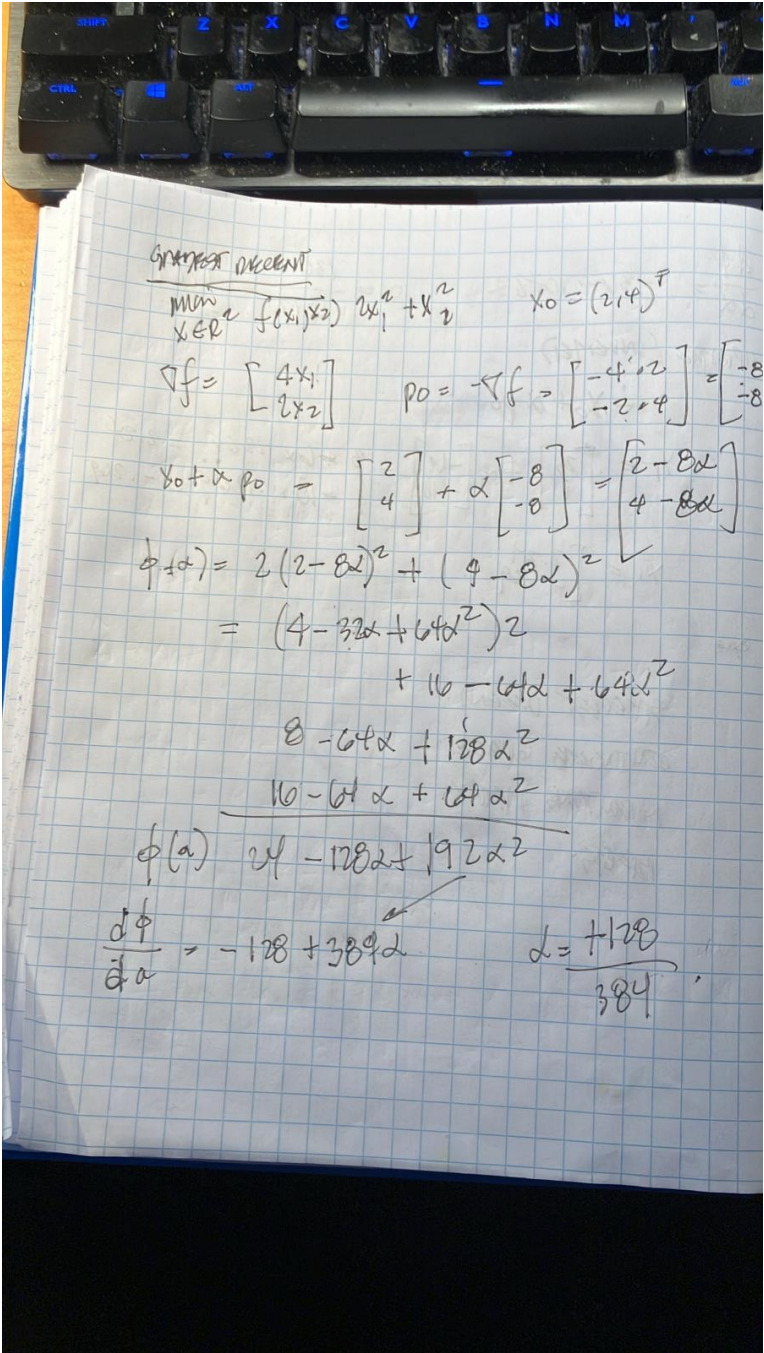


3. Mutation



Adv: global search; no gradient; handles discrete/mixed/discontinuous f

Dis: many f evals; no convergence guarantee; tuning required (p_c, p_m , pop size)



$x_1, x_2, \dots, x_n$	components of design variable vector
$X_0, X_1, \dots, X_k$	sequence of iterates
$\hat{X}_k$	candidate point (α still unknown)
p_k	search direction at iteration k
α_k	step size at iteration k (line search)
$\phi(\alpha)$	$f(\hat{X}_k)$ — 1D slice along search line
$\nabla f(X_k)$	gradient evaluated at X_k
$H_f(X_k)$	Hessian evaluated at X_k
$X_k = X_{k-1} + \alpha_{k-1} p_{k-1}$	— general update

1. $\nabla f(X_{k-1})$
2. $p_{k-1} = -\nabla f$
3. $\hat{X}_k = X_{k-1} + \alpha p_{k-1}$
4. $\phi(\alpha) = f(\hat{X}_k)$
5. solve $d\phi/d\alpha = 0$ for α_{k-1}
6. $X_k = X_{k-1} + \alpha_{k-1} p_{k-1}$

$$\beta_{k-1} = \frac{\|\nabla f(X_{k-1})\|^2}{\|\nabla f(X_{k-2})\|^2}$$
$$\beta_{k-1} = \frac{g_k^\top (g_k - g_{k-1})}{\|g_{k-1}\|^2}$$

Condition	Statement	Meaning
1st-order necessary	$\nabla f(X^*) = 0$	stationary point
2nd-order necessary	$H_f(X^*) \succeq 0$	no downward curvature
Sufficient (min)	$\nabla f = 0$ and $H_f \succ 0$	confirmed minimum
$H_f \succ 0 = \text{min}; \quad H_f \prec 0 = \text{max}; \quad \text{indefinite} = \text{saddle}.$		

1. $\nabla f(X_{k-1})$
2. $H_f(X_{k-1})$, invert
3. $p_{k-1} = -H_f^{-1}\nabla f$
4. try $\alpha = 1$: $\hat{X}_k = X_{k-1} + p_{k-1}$
5. line search if needed
6. $X_k = X_{k-1} + \alpha_{k-1}p_{k-1}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\nabla f(X_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}, \quad H_f^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

$X_1 = (0, 0)^\top$, $f(X_1) = 0$ — one step exact

1. $p = -\tilde{V}_{k-1}\nabla f$; **l.s.** $\rightarrow \alpha_{k-1}$; $X_k = X_{k-1} + \alpha p$
2. $s = X_k - X_{k-1}$, $y = \nabla f(X_k) - \nabla f(X_{k-1})$, $\sigma = 1/(y^\top s)$
3. $\tilde{V}_k = (I - \sigma s y^\top)\tilde{V}_{k-1}(I - \sigma y s^\top) + \sigma s s^\top$

Worked: $X_0 = (3, 1)^\top$, $\nabla f_0 = (6, 10)^\top$
 $\alpha_0 \approx 0.127$; $X_1 = (2.24, -0.27)^\top$
 $s_0 = (-0.76, -1.27)^\top$; $y_0 = (-1.52, -12.69)^\top$

Method	Direction p_{k-1}
SD	$-\nabla f$
CG	$-\nabla f + \beta p_{k-2}$
Newton	$-H_f^{-1} \nabla f$
BFGS	$-\tilde{V}_{k-1} \nabla f$

Constrained standard form: $\min f(X)$ s.t. $h_j(X) = 0, \ g_i(X) \leq 0$

Lagrangian: $\mathcal{L} = f(X) + \sum_j \lambda_j h_j + \sum_i \mu_i g_i$

KKT conditions (necessary; sufficient for convex):

1. Stationarity: $\nabla_X \mathcal{L} = 0$
2. Primal feasibility: $h_j(X^*) = 0, \ g_i(X^*) \leq 0$
3. Dual feasibility: $\mu_i^* \geq 0$
4. Complementary slackness: $\mu_i^* g_i(X^*) = 0$
(inactive constraint $\Rightarrow \mu_i^* = 0$)

KKT analytical $\min x_1^2 + 5x_2^2$ s.t. $x_1 + x_2 \geq 4$

Rewrite: $g = 4 - x_1 - x_2 \leq 0$. Test $X_{\text{unc}} = (0, 0)$: $g = 4 > 0$ — infeasible $\Rightarrow$ active.

$l = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$

$\partial l / \partial x_1 = 2x_1 - \lambda = 0 \Rightarrow x_1 = \lambda / 2$

$\partial l / \partial x_2 = 10x_2 - \lambda = 0 \Rightarrow x_2 = \lambda / 10$

Sub into $x_1 + x_2 = 4$: $\lambda / 2 + \lambda / 10 = 4 \Rightarrow \lambda = 20 / 3$

$x^* = (10 / 3, \ 2 / 3)^\top, \quad f^* = 40 / 3$

Residual and Jacobian (shared with Newton on KKT):

$$r = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix}$$

$$J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

Newton on KKT init $z_0 = (0, 0, 0)^\top$

$r(z_0) = (0, 0, -4)^\top$; solve $J \Delta z = -r(z_0) = (0, 0, 4)^\top$:

Row 1: $\Delta x_1 = \Delta \lambda / 2$ Row 2: $\Delta x_2 = \Delta \lambda / 10$

Row 3: $\frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6 \Delta \lambda = 40 \Rightarrow \Delta \lambda = 20 / 3$

$\Delta x_1 = 10 / 3, \ \Delta x_2 = 2 / 3$

$z_1 = z_0 + \Delta z = (10 / 3, \ 2 / 3, \ 20 / 3) = x^*$

(one step — KKT system is linear here)

Penalty method $\min x_1^2 + 5x_2^2 + \frac{c}{2}(x_1 + x_2 - 4)^2$

$\partial / \partial x_1$: $2x_1 + c(x_1 + x_2 - 4) = 0$

$\partial / \partial x_2$: $10x_2 + c(x_1 + x_2 - 4) = 0$

$\Rightarrow x_1 = 5x_2$; as $c \rightarrow \infty$: $x^* \rightarrow (10 / 3, \ 2 / 3)$

Each c gives an unconstrained subproblem.

Penalty: $+\frac{c}{2} \sum \max(0, g_i)^2$; increase c

Int. pt.: barrier $-\mu \sum \log(-g_i)$; decrease $\mu \rightarrow 0$

SQP: QP subproblem at each iterate (Newton on KKT)

LP standard form: $\min c^\top x$ s.t. $Ax = b, \ x \geq 0$

Four steps to convert:

- 1. **Objective:** if max, negate
- 2. **Free variable:** $x = x^+ - x^-$, $x^+, x^- \geq 0$
- 3. **$\leq$ constraint:** add slack $s \geq 0$
- 4. **$\geq$ constraint:** subtract surplus $s \geq 0$

(a) [5 marks] Convert to standard LP form (minimise, equality constraints, $x \geq 0$):

$\max 2x_1 - x_2 + 8x_3 \quad \text{s.t.} \quad 7x_1 + 5x_2 - x_3 \leq 1, \quad x_1 + 6x_2 \geq 4, \quad x_1, x_2 \geq 0, \ x_3 \text{ free.}$

Solution: Four steps:

- 1. **Objective:** negate $\Rightarrow \min -2x_1 + x_2 - 8x_3$
- 2. **Free variable:** $x_3 = x_3^+ - x_3^-$, $x_3^+, x_3^- \geq 0$
- 3. **$\leq$ constraint:** add slack $s_1 \geq 0$: $7x_1 + 5x_2 - x_3^+ + x_3^- + s_1 = 1$
- 4. **$\geq$ constraint:** subtract surplus $s_2 \geq 0$: $x_1 + 6x_2 - s_2 = 4$

Standard form:

$$\min -2x_1 + x_2 - 8x_3^+ + 8x_3^- \quad \text{s.t.} \quad \underbrace{\begin{bmatrix} 7 & 5 & -1 & 1 & 1 & 0 \\ 1 & 6 & 0 & 0 & 0 & -1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ x_3^+ \\ x_3^- \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \text{all variables} \geq 0.$$

Problem classification:

Type	Key feature
LP	f and all constraints linear
QP	f quadratic, constraints linear
NLP	nonlinear f or constraints
Convex	f convex, feasible set convex
Non-convex	multiple local minima possible
Unconstrained	no h, g
Equality constrained	$h_j = 0$ only
Inequality constrained	$g_i \leq 0$ included

MPC — receding horizon control:

Finite-horizon problem at each step k :

$$\min_{u_0, \dots, u_{N-1}} \sum_{t=0}^{N-1} c(x_t, u_t) + V_f(x_N)$$

s.t. $x_{t+1} = f(x_t, u_t), \ x_t \in \mathcal{X}, \ u_t \in \mathcal{U}$

Apply u_0^* only; advance; repeat.

Dynamic programming: $V_N = V_f(x); \quad V_k(x) = \min_u [c(x, u) + V_{k+1}(f(x, u))]$

Bellman: sub-sequences of an optimal sequence are optimal.

Adv: hard constraints; optimal traj; short horizons

Disadv: needs model; computationally expensive; stability hard